IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2016

MSc Control Systems

Corrected copy

GAME THEORY

Wednesday, 27 April 10:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : R.B. Vinter
Second Marker(s) : D. Angeli

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Information for candidates:

No special instructions for candidates.

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1. Consider the zero-mean matrix game with pay-off matrix

$A\backslash B$			
	6	0	7
	5	6	0
	6	4	4

in which the A-player (B-player) aims to minimize (maximize) the pay-off.

(i): Show that the last row is dominated by a mixture of the first rows. [2]

(ii): Does the this matrix game (in pure strategies) possess the strong Minimax property? Deduce from your answer whether, or not, there exists a saddlepoint equilibrium in pure strategies. [4]

(iii): Determine, by graphical or other methods, a saddle-point equilibrium of the game in mixed strategies. [14]

Hint: Use the results of part (i) to reduce the pay-off matrix to a 2×3 *matrix.*

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2. Consider the extensive form, non-zero sum game, involving the 3 players (Players A, B and C), described by the diagram.

(Interpretation: Players A, B and C are users of a communications network, who choose between two service providers, one or which is fast and unreliable, and the other of which is slow and reliable, to minimize connection time.)

Players A, B and C can each select F or S. (F = 'fast', M =, S = 'slow'). The three numbers at the end of each branch are the pay-offs (to be minimized) for Player A, Player B and Player C, in that order.

Note the indicated information set. Player A selects a strategy first (and has no knowledge of the other players selections). Player B selects a stategy second, knowing whether Player A has selected 'F' or 'S'. Player C chooses last (without knowing whether Player A has chosen 'F' or 'S').

Player A selects the mixed strategy

$$\text{Player } A \text{ selects} \begin{cases} F & \text{with probability } 0.25, \\ S & \text{with probability } 0.75 \end{cases} \tag{1}$$

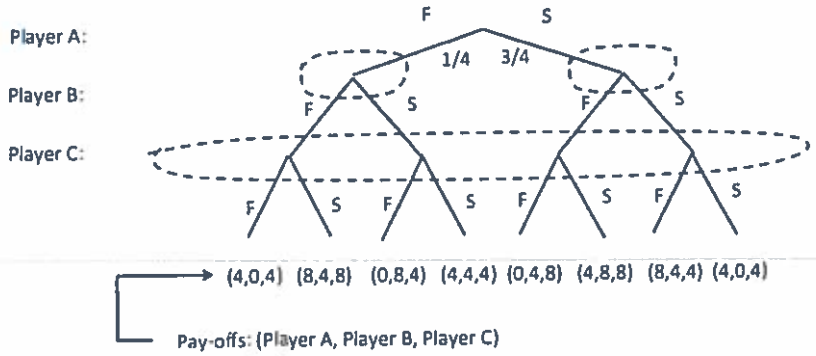
and informs of Players B and C that this is his strategy.

(i): Calculate the average pay-off matrices for the three players. [7]

Determine a Nash equilibrium for Players B and C (in pure strategies), based on the pay-off matrices for these two players alone. [8]

Determine the pay-offs for the three players for their chosen strategies. [2]

(ii): Without doing any calculations, comment on whether you think player B has achieved a better pay-off as a result of his 'extra' knowledge: Player A has chosen 'F' or 'S'. [3]



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3. N people ('players'), at a remote location, have mobile phones, which thay can use to send a text message over a low-capacity communication channel. If only one person sends a text, the message is transmitted. But if two, or more, people attempt to send a message, the communication channel fails and so the messages are not transmitted. Each person must decide whether to send a text ('send') or not send a text ('not send').

For player i, the 'satisfaction' pay-offs (which all players aim to maximize) are as follows:

Pay-off (for player i) = 2 if player i chooses 'not send'.

Pay-off (for player i) = 3 if player i chooses 'send', and no other player chooses 'send'.

Pay-off (for player i) = 1 if player i chooses 'send', and at least one other player chooses 'send'.

(i): Determine a symmetric, randomized, independent strategy for the N players, of the form:

Player i chooses 'send' w.p. $\bar{\alpha}(N)$ and chooses 'not send' w.p. $(1-\bar{\alpha}(N))$

for player $i = 1, \ldots, N$, such that these randomize strategies give a Nash equilibrium over mixed strategies. (Your formula for $\bar{\alpha}(N)$ will depend on N.) [15]

(ii): Determine

$$P(N) := \{\text{probability that nobody chooses 'send'}\}. \quad [3]$$

Sketch the graph of $P(N)$ as N increases, and comment on the nature of the graph. [2]

Hint: Use the fact that, if $N-1$ players independently choose 'send' with probability $\bar{\alpha}$, then the probability that one or more of these players will choose 'send' is $1-(1-\bar{\alpha})^{N-1}$.

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4. Two companies, A and B, independently select prices for their products, $p_A\,(\geq 0)$ and $p_B\,(\geq 0)$ respectively. Company A also spends $a(>0)$ on advertising the products. Consumer demand for company A's product, d_A, and for company B's product, d_B, are modelled (modified Cournot duopoly model) as

$$\begin{cases} d_A = (a - p_A - p_B) \\ d_B = (a - p_A - p_B)\,. \end{cases}$$

The profits of company A (excluding the cost of advertising) are $d_A \times p_A$, and the profits of company B are $d_B \times p_B$. Company B (and, of course, also Company B) knows the level a of Company A's advertising. Each company seeks to maximize its profit.

(i): Assume that a is fixed. Assume also that Company A sets its price without knowing Company B's price, and Company B sets its price without knowing Company B's price.

Draw the response curves for the two companies. [4]

Find the Nash equilibrium, and evaluate the pay-offs. [6]

Now suppose that Company B knows Company A's price when company B's price is set. Regarding Company A as the leader and Company B as the follower, calculate the Stackelberg strategies and evaluate the pay-offs. [6]

(ii) Suppose

$$\text{Company A's advertising cost} = (1/3)a^3\,.$$

(The cost of advertising must be subtracted from Company A's profits.) Using the pay-off formulae you obtained in part (i), determine the level of advertising $\bar{a}$, that will maximize the profits of Company A, in both cases when Company B knows, and does not know, Company A's prices. [4]

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5. Consider the two-stage, scalar, dynamic zero-sum game, in which the u_0 control and the v_0 control are chosen to minimize and maximize the pay-off, respectively.

$$\begin{cases} \underset{u_0}{\text{Min}}\ \underset{v_0}{\text{Max}}\ x_1^2 + u_0^2 - \alpha v_0^2 \\ \text{with dynamic constraint:} \\ \qquad\qquad x_1 = x_0 + u_0 + v_0\,. \end{cases}$$

Here, $\alpha > 1$ is a constant and x_0 ('the initial state') is a given number.

(i): Derive a formula for the pay-off $J_\alpha(u_0, v_0; x_0)$. (It will depend on the controls u_0 and v_0, initial state x_0, and also the parameter α). [2]

(ii): Derive formulae for the saddle point equilibrium of the two players $(\bar{u}_0(\alpha, x_0), \bar{v}_0(\alpha, x_0))$ and the value of the pay-off at equilibrium. (They will depend on α and x_0.) *(You may assume that a saddle point exists and is unique.)* [12]

(iii): Fix x_0. Show that, in the limit as $\alpha \to \infty$,

$$(\bar{u}_0(\alpha, x_0), \bar{v}_0(\alpha, x_0)) \quad \to \quad (u_0^*(x_0), 0) \qquad [3]$$

and

$$J_\alpha(\bar{u}_0(\alpha, x_0), \bar{v}_0(\alpha, x_0); x_0) \quad \to \quad J_0(u_0^*(x_0), 0; x_0), \qquad [3]$$

where $u_0^*(x_0)$ minimizes $J_0(u_0, 0; x_0)$ over u_0, i.e. $u_0^*(x_0)$ is the minimizer of the pay-off with respect to u_0, when the v_0 control is fixed at $v_0 = 0$.

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